

Robot-assisted Neural Radiance fields for plot-level cotton crop 3D reconstruction and yield estimation

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ABSTRACT

Cotton boll counting is a critical component of yield estimation and plays a vital role in advancing plant phenomics and precision agriculture. However, the complex spatial distribution of cotton bolls and frequent occlusions in two-dimensional (2D) images pose significant challenges to accurate quantification. To address these issues, we developed a novel workflow based on Neural Radiance Fields (NeRF), which integrates high-throughput robotic image data acquisition, three-dimensional (3D) reconstruction, and cotton boll mapping at the plot level in a breeding field. A customized robotic platform equipped with ten cameras mounted around the canopy frame was used to capture overlapping red, green, and blue (RGB) images of plot-level cotton plants. NeRF was then employed to reconstruct detailed 3D models, and the reconstruction quality was evaluated. Furthermore, this study presents a comprehensive 2D-to-3D mapping approach based on an improved FruitNeRF algorithm, which effectively overcomes the challenges of 3D plot-level cotton boll annotation and limited training data. The proposed approach enables accurate quantification of cotton boll numbers and provides a reliable foundation for yield prediction. Experimental results demonstrated that the Lumix-camera configuration mounted on the robotic platform achieved the best overall reconstruction performance among the tested configurations, with a peak signal-to-noise ratio (PSNR) of 17.21 dB, a structural similarity index (SSIM) of 0.49, and a learned perceptual image patch similarity (LPIPS) of 0.54. This configuration was also better suited for high-throughput plot-level field phenotyping because it enabled more stable and standardized image acquisition. The reconstructed plot-level point clouds retained sufficient structural information for downstream cotton boll segmentation, counting, and yield estimation. In the exported plot-level point clouds, both cotton bolls and branches are clearly distinguishable. Based on Lumix data, the mean absolute percentage error (MAPE) for cotton boll number and boll weight prediction was 5.43% and 8.78%, respectively. Overall, this approach enables high-throughput phenotyping and the generation of plot-level cotton point clouds suitable for downstream boll counting and yield estimation, providing a valuable framework for plot-level cotton phenotyping and precision management for low-density, post-defoliated plots under the experimental conditions evaluated in this study.

1. Introduction

Cotton (*Gossypium* spp.) is the most important renewable fiber crop globally, with an annual production value exceeding USD 46 billion, supporting the livelihoods of 150 million cotton workers across 75 countries (Devoto et al., 2024; Kun et al., 2025). As the global population continues to rise, improving cotton yield has become a critical

challenge in modern agriculture. Plant phenotypic information plays a central role in breeding programs by providing essential data to support yield enhancement (Conaty et al., 2022; Voss-Fels et al., 2019). Among these traits, boll number is widely recognized as one of the most critical indicators of cotton yield in breeding programs. It offers valuable insights into genotype performance under diverse environments and serves as a key criterion for selecting high-yielding lines and guiding

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genetic improvement (Virk et al., 2024). However, current boll counting in cotton predominantly relies on manual field sampling, which is time-consuming, labor-intensive, and susceptible to human error, particularly under high-temperature field conditions (Chen et al., 2025). These limitations underscore the urgent need for efficient, accurate, and reliable approaches to boll number quantification.

In cotton breeding and field phenotyping, accurate characterization of boll distribution and plant architecture at the plot level remains a significant challenge under field conditions. Cotton bolls are spatially distributed throughout dense and partially occluded canopies, making reliable quantification difficult using conventional manual inspection or 2D image-based approaches. Manual field scoring is labor-intensive, subjective, and difficult to scale to large breeding trials. Unmanned Aerial Vehicles (UAV)-based imaging systems primarily rely on top-view observations and often fail to capture detailed side-view and under-canopy structures that are important for understanding boll distribution and plant architecture. In addition, conventional 2D image-based methods are inherently viewpoint-dependent and are susceptible to occlusion and overlapping cotton bolls, particularly in dense field scenes. These limitations motivate the need for semantic 3D reconstruction methods capable of generating detailed plot-level cotton representations for downstream phenotyping and yield-related analysis.

Two-dimensional (2D) imaging techniques combined with deep learning have been widely applied to cotton boll counting and to fruit detection tasks in various crops (Wang et al., 2025). In this context, some researchers have investigated segmentation methods based on both fully supervised and weakly supervised learning to evaluate boll counting accuracy using proximal imagery (Adke et al., 2022). Aerial images have also been used to predict cotton boll number using density map-based regression approaches, such as DM-Count (Tan et al., 2025). The 2D detection-based method has also been applied to fruit detection in many other crops. For example, YOLOv5x has demonstrated outstanding performance in kiwifruit detection, achieving an average precision of 99.5% (Gao et al., 2024), while YOLOv8n achieved an average accuracy of 91.93% in strawberry identification (An et al., 2024). However, fruits with complex spatial arrangements, such as blueberries, grapes and apples, present greater challenges for comprehensive detection in 2D images due to occlusion issues (Ni et al., 2021; Zheng et al., 2024; Yang et al., 2025). Cotton bolls are also distributed in a three-dimensional (3D) space on the plant, particularly at the plot level, where severe occlusions in 2D images significantly compromise counting accuracy.

Another major challenge in 3D plant phenotyping is the requirement for extensive manual annotation of 3D point clouds for semantic segmentation and structural analysis. Compared with 2D image annotation, dense point-level annotation in complex plant canopies is highly labor-intensive and time-consuming. To address this limitation, recent advances in foundation models and semantic Neural Radiance Fields (NeRF) frameworks provide opportunities to transfer semantic information from 2D image space into 3D representations without requiring large-scale manually labeled 3D point cloud datasets.

Compared to 2D images, point clouds preserve the 3D spatial architecture of plants, which helps mitigate occlusion issues and enables detailed 3D analysis of plant traits. Building on this advantage, recent studies have employed point cloud instance segmentation to extract a variety of phenotypic traits, such as branch length and canopy volume in apple trees, and plant height and leaf angle in maize (Jiang et al., 2025a; Miao et al., 2021). In fruit phenotyping, segmenting individual apples within point clouds enables accurate fruit counting and volume estimation (Dong et al., 2023). High-precision 3D point clouds are a prerequisite for successful segmentation. Terrestrial Light Detection and Ranging (LiDAR) systems can directly acquire high-quality point clouds; however, their high cost still limits their widespread use in practice. By contrast, RGB-D cameras provide a more cost-effective alternative, but their limited resolution often results in point clouds with substantial loss of fine-scale detail (Bloch et al., 2025). Therefore, developing an efficient, low-cost, and high-fidelity 3D reconstruction method is critical for

advancing plant phenotyping and trait analysis (Jiang et al., 2024). An important practical advantage of the proposed workflow is that it enables plot-level 3D reconstruction and cotton boll analysis directly from RGB image inputs collected under field conditions, without relying on specialized depth sensors or LiDAR systems.

Different sensing platforms present distinct trade-offs for field-based cotton phenotyping. UAV-based imaging systems are well-suited for large-area field monitoring and high-throughput scouting; however, their predominantly top-view observations limit their ability to capture detailed side-view and under-canopy structures in dense cotton canopies. Handheld smartphone acquisition provides flexibility and relatively low hardware cost, but suffers from reduced repeatability and greater operator-dependent variability during large-scale field phenotyping. Terrestrial LiDAR systems can directly generate highly accurate point clouds with rich geometric detail; however, they generally do not provide color and texture information of plants. In addition, their high equipment cost and limited scalability remain challenges for practical breeding applications.

In this study, a ground robotic platform equipped with multiple RGB cameras was selected to balance acquisition cost, reconstruction quality, scalability, and repeatability for plot-level cotton phenotyping. Compared with UAV-based systems, the ground robot enables more stable multi-view acquisition of side-view and under-canopy structures that are important for cotton boll distribution and architectural analysis. In addition, compared with handheld acquisition, the robotic platform provides more consistent imaging trajectories and standardized multi-view observations across plots.

Plot-level phenotyping was selected as the focus of this study because cotton breeding and field phenotyping are commonly conducted at the plot scale, where multiple plants representing the same genotype are evaluated collectively for yield-related traits and architectural characteristics. Compared with single-plant reconstruction, plot-level analysis introduces additional challenges, including overlapping plant structures, increased occlusion, and more complex spatial distributions of cotton bolls under field conditions. Therefore, developing semantic 3D reconstruction methods for plot-level cotton analysis is important for practical breeding-oriented phenotyping applications.

Multi-view 3D reconstruction has emerged as a promising approach for achieving a balance between sensor cost and model resolution. This approach reconstructs the 3D structure of objects or scenes by processing multiple overlapping 2D images captured from different viewpoints. The traditional multi-view 3D reconstruction pipeline typically involves two distinct stages: the Structure-from-Motion (SfM) algorithm enables the recovery of both camera trajectories and the 3D structure of a scene from a collection of partially overlapping 2D images (Schönberger and Frahm, 2016). The process begins with feature extraction, followed by matching corresponding features across overlapping image pairs to establish image correspondences. Based on these correspondences, the three-dimensional positions of matched features—forming a sparse point cloud—can be recovered through an iterative bundle adjustment (BA) algorithm (Eltner et al., 2016). Subsequently, a multi-view stereo (MVS) algorithm is applied to recover the depth of each pixel in each image using consistent matches across views, resulting in a dense point cloud (Meinen and Robinson, 2020; Smith et al., 2016). Owing to its robustness and compatibility with consumer-grade cameras, this classical workflow has been widely adopted across a broad range of object- and scene-level reconstruction tasks. It has demonstrated strong performance in plant phenotyping applications, including the 3D modeling of individual plants such as Carex and kale (Yang et al., 2024a). A variety of software platforms that integrate these workflows—such as COLMAP, Pix4D mapper (Pix4D S.A., Lucerne, Switzerland), and Agisoft Metashape (Agisoft LLC, St. Petersburg, Russia)—are capable of reconstructing 3D object models from multiple overlapping 2D images. A variety of agricultural studies have adopted these tools for 3D reconstruction tasks, successfully modeling crops such as cotton and maize (Sun et al., 2020; Wu et al., 2020). Although SfM-MVS-based 3D

reconstruction methods can perform well under controlled conditions, their accuracy often deteriorates in field environments due to challenges such as dense foliage, severe occlusions, and variable illumination (James et al., 2025; Jiang et al., 2025c). These difficulties are exacerbated by the fragmented nature of the traditional SfM-MVS pipeline, which involves multiple discrete stages and increases the risk of reconstruction failure. Together, these limitations highlight the need for more robust and integrated reconstruction approaches.

NeRF have emerged as a novel learning-based approach that leverages neural networks to implicitly represent 3D scenes through a continuous volumetric radiance field (Mildenhall et al., 2020). The NeRF method models the color and density at any 3D location conditioned on a viewing direction using a multi-layer perceptron (MLP). The network is trained to predict view-dependent color and volume density from five inputs (spatial position and viewing direction) by minimizing the pixel-wise reconstruction error between rendered and ground-truth images. This process enables the implicit encoding of scene geometry and appearance directly from 2D images. By circumventing traditional geometric components such as MVS, surface reconstruction, and texture mapping, NeRF simplifies the 3D reconstruction pipeline into an end-to-end optimization framework. Once trained, NeRF can synthesize novel views with photorealistic quality, even for scenes with complex lighting and fine-grained structures. NeRF has been successfully applied to the high-fidelity reconstruction of various fruits, including dragon fruit, grapes, lychees, and oranges (Hu et al., 2024). This method supports not only the 3D reconstruction of individual fruits but also entire plants. Moreover, its feasibility has been demonstrated at the plot level for crops such as corn and strawberries (Arshad et al., 2024; Zhang et al., 2024). Although recent years have witnessed notable progress, the intricate architecture of cotton plants and the dense distribution of bolls pose significant challenges, highlighting the need for further exploration of NeRF's 3D reconstruction capabilities at the plot-level. In particular, integrating robotic platforms for high-throughput data acquisition and ensuring high-fidelity plot-scale reconstruction remain underexplored yet critical directions.

Following the reconstruction of plot-level 3D models of cotton plants, the segmentation of cotton bolls is a critical prerequisite for accurate boll counting. To achieve this, two potential approaches can be employed for 3D cotton boll segmentation. One common method involves exporting point clouds generated by the NeRF model, which can then be used for various point cloud processing tasks. For instance, PointNet++ (Qi et al., 2017) has been successfully applied to segment bolls from the point cloud of cotton plants (Jiang et al., 2022). However, this approach requires extensive annotation of point clouds, leading to significant time spent on data labeling, particularly when labeling plants at the plot-level, which is even more challenging. Additionally, the scarcity of 3D point cloud data is also a notable issue. Another method leverages multi-view 3D reconstruction techniques to establish a connection between 2D and 3D. The process first obtains 2D masks of the target from 2D images, then projects these masks into 3D space, and finally segments the object in the 3D space. This processing pipeline has already been successfully applied to fruits such as apples and blueberries (Gené-Mola et al., 2020; Ni et al., 2021), as well as to crops such as rice panicles (Yang et al., 2024b). Combining NeRF with object counting enables the embedding of 2D semantic information into 3D space, resulting in a Semantic Neural Radiance Field that directly outputs target-specific point clouds. This approach not only eliminates the need for training an additional 3D segmentation model, but also circumvents the challenges associated with manual 3D annotation and the scarcity of high-quality 3D training data. A key methodological advantage of the proposed workflow is that it reduces the dependence on extensive manually annotated 3D point cloud datasets by leveraging semantic 2D-to-3D mapping through foundation-model-assisted segmentation and semantic NeRF reconstruction. Semantic-NeRF extends NeRF by enabling the joint learning and representation of scene geometry, appearance, and semantic information (Zhi et al., 2021). FruitNeRF is a

NeRF-based model designed for fruit-centric semantic perception. It has been integrated into the Nerfstudio framework (Tancik et al., 2023), enabling simultaneous 3D reconstruction and fruit segmentation from multi-view RGB images (Meyer et al., 2024). Despite its promising potential for plot-level cotton analysis, this integration of NeRF with 3D segmentation remains largely unexplored in existing research.

The overall goal of this study was to utilize a customized multi-camera robotic platform for multi-view data acquisition and to employ advanced Neural Radiance Fields (NeRF) for plot-level 3D reconstruction and segmentation of cotton plants. A novel method was developed for 3D cotton boll segmentation, counting, and yield estimation by projecting 2D cotton boll masks into 3D space. The specific objectives of this research were to: (1) design an innovative data acquisition method using a ground robot equipped with multiple cameras, and to compare its performance with that of a handheld smartphone; (2) perform plot-level 3D reconstruction of cotton plants using the NeRF method and evaluate the reconstruction quality; (3) develop a 2D annotation strategy based on the Segment Anything Model (SAM) for constructing a cotton boll segmentation dataset; and (4) implement plot-level cotton boll counting and yield estimation based on 3D segmentation.

2. Materials and methods

2.1. Experimental field setup and data collection

The data used in this experiment were collected on October 25, 2024, in Tifton, Georgia, USA (Coordinates: 31.4371° N, 83.5855° W, Fig. 1a). The experimental field consisted of 160 plots (Fig. 1b), comprising 32 different genotypes, with each plot containing a single genotype. The plant height of these genotypes ranges around 1 m. The center point of each plot was used as a reference, with a row spacing of 6 feet and a column spacing of 25 feet (Fig. 1e). Each plot measured approximately 18 feet in length. A defoliant was applied to the entire field one week prior to data collection. In some plots, most of the leaves had already fallen off (Fig. 1c), while others still retained a substantial amount of foliage (Fig. 1d). The selected experimental conditions, including post-defoliation and relatively low planting density, were intended to provide an initial validation scenario for the proposed semantic 3D reconstruction workflow under field conditions. Although partial canopy occlusions and structural complexity were still present, these conditions are less challenging than fully closed, high-density commercial cotton canopies. Nearly all bolls had already opened, with only a few remaining green. These immature bolls were excluded from yield measurements in this study. Data for testing were collected from the first, second, and seventh rows (from west to east).

Two data collection methods were employed in this experiment. The first was a high-throughput approach utilizing a ground robot equipped with multiple cameras, and the second was a low-cost method based on a smartphone. The data collection was conducted on a sunny day with clear skies. The smartphone data was collected between 11:00 AM and 2:00 PM Eastern Time, while the ground robot collected data between 4:00 PM and 6:00 PM. The ground robot, MARS (Xu and Li, 2022), was equipped with ten cameras, which were mounted approximately evenly around the canopy bracket (Fig. 2a). The two vertical blue support bars that hold the camera mounting frame are spaced approximately 2.48 m apart. The horizontal blue bar, parallel to the ground and used for camera installation, is positioned at a height of approximately 1.97 m above the ground. Due to limited camera availability, two different camera types were used on MARS. Due to the experimental setup used in this study, the robot was operated using manual navigation during data collection rather than autonomous navigation. During field data collection, the MARS robot was manually operated to move between cotton rows to ensure stable multi-view image acquisition across different plots. The scanning speed during data collection was approximately 0.15–0.2 m/s. The cameras were mounted at multiple viewing directions around the cotton canopy, and the camera mounting angles

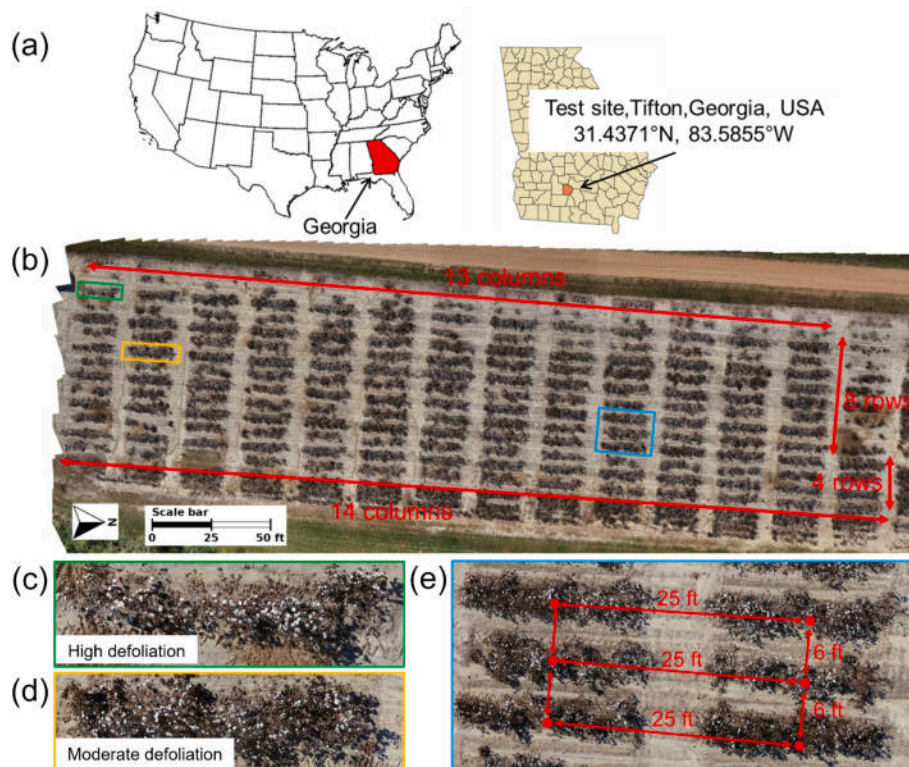


Fig. 1. Overview of the experimental field. (a) Location of the data collection site. (b) Plot-level planting layout of cotton plants in the experimental field. (c) and (d) show two example plots with different levels of defoliation. In (c), most of the leaves have fallen off, making the plot appear relatively sparse. In contrast, (d) shows a plot where many leaves are still retained, and the plants also appear denser. (e) Red dots represent the center points of individual plots. The spacing between the centers of adjacent plots is 6 ft in the row direction and 25 ft in the column direction.

were manually adjusted to maximize canopy coverage and maintain the cotton plants near the center of the image frame during scanning. In Fig. 2b, the blue circles indicate the positions of five Lumix DMC-G7 cameras (Panasonic Corporation, Osaka, Japan), and the red circles represent five Fuji X-A10 cameras (Fujifilm Holdings Corporation, Tokyo, Japan). All cameras recorded video at 30 frames per second (FPS) for plant scanning. During video recording, the Lumix and Fuji cameras were set to automatic mode (Table 1). Lumix cameras recorded at 3840×2160 resolution, while Fuji cameras recorded at 1920×1080 . To maximize the field of view, the focal length of each camera was set to its minimum, and the camera angles were fine-tuned to center the plants within the image frame. On average, scanning a single plot took approximately 40 s, and a total of 39 plots were scanned using this method.

Video collection using a smartphone offers greater operational flexibility and cost-effectiveness, making it suitable for sampling-based data collection in field environments. The smartphone was mounted on a handheld stabilizer and used to scan the plants directly. Videos were recorded at a resolution of 3840×2160 and a frame rate of 30 FPS. An ultra-wide-angle lens was used throughout smartphone-based data collection to maximize the available field of view. The specific data acquisition path is illustrated in Fig. 2c, where the top view was captured twice, and the side view was acquired by orbiting the plant at two different height levels. On average, scanning a single plot using the smartphone took approximately 130 s, with a total of 41 plots scanned using this method. There are 26 identical plots captured using two different data acquisition methods. These plots are used in the subsequent process for evaluating cotton boll counting accuracy. The remaining plots are used for training the 2D segmentation model. Table 1 summarizes the specific parameters used for data acquisition.

2.2. Plot-level cotton plant 3D reconstruction based on NeRF

Neural Radiance Fields (NeRF) can be briefly summarized as using a neural network to implicitly learn a static 3D scene, enabling 3D reconstruction of complex environments and novel view synthesis from arbitrary perspectives (Mildenhall et al., 2020). The input to NeRF consists of a 3D coordinate (x, y, z) and a viewing direction (θ, ϕ), while the output is the color (R, G, B) and volumetric density (σ) at that point in the specified direction. By sampling points along camera rays and applying volume rendering techniques, NeRF can synthesize photo-realistic images from arbitrary viewpoints using multiple 2D images, while preserving detailed geometric and texture information.

NeRF was employed to reconstruct cotton plants at the plot-level and to generate corresponding point clouds in this study. Due to differences in video resolution across cameras, images were extracted from the recorded videos and uniformly resampled to a resolution of 1920×1080 . For plot-level cotton videos collected using multiple cameras, approximately 90 frames were uniformly sampled from each video, while 500 frames were extracted from the video captured with the smartphone. In both cases, the sampling strategy ensured more than 90% overlap between adjacent frames. During frame extraction, the overlap between adjacent sampled frames was estimated based on the displacement of the imaging system between consecutive sampled frames relative to the approximate image coverage along the scanning direction. Given the robot scanning speed of approximately 0.15–0.2 m/s, the video frame rate of 30 FPS, and the scanning duration of approximately 40 s per plot for each camera, approximately 90 frames were uniformly extracted from each camera video. This sampling strategy was expected to preserve more than 90% overlap between adjacent sampled frames. The overlap was approximately estimated from the acquisition speed, frame rate, and image coverage, rather than being calculated using detailed image registration. This high-overlap

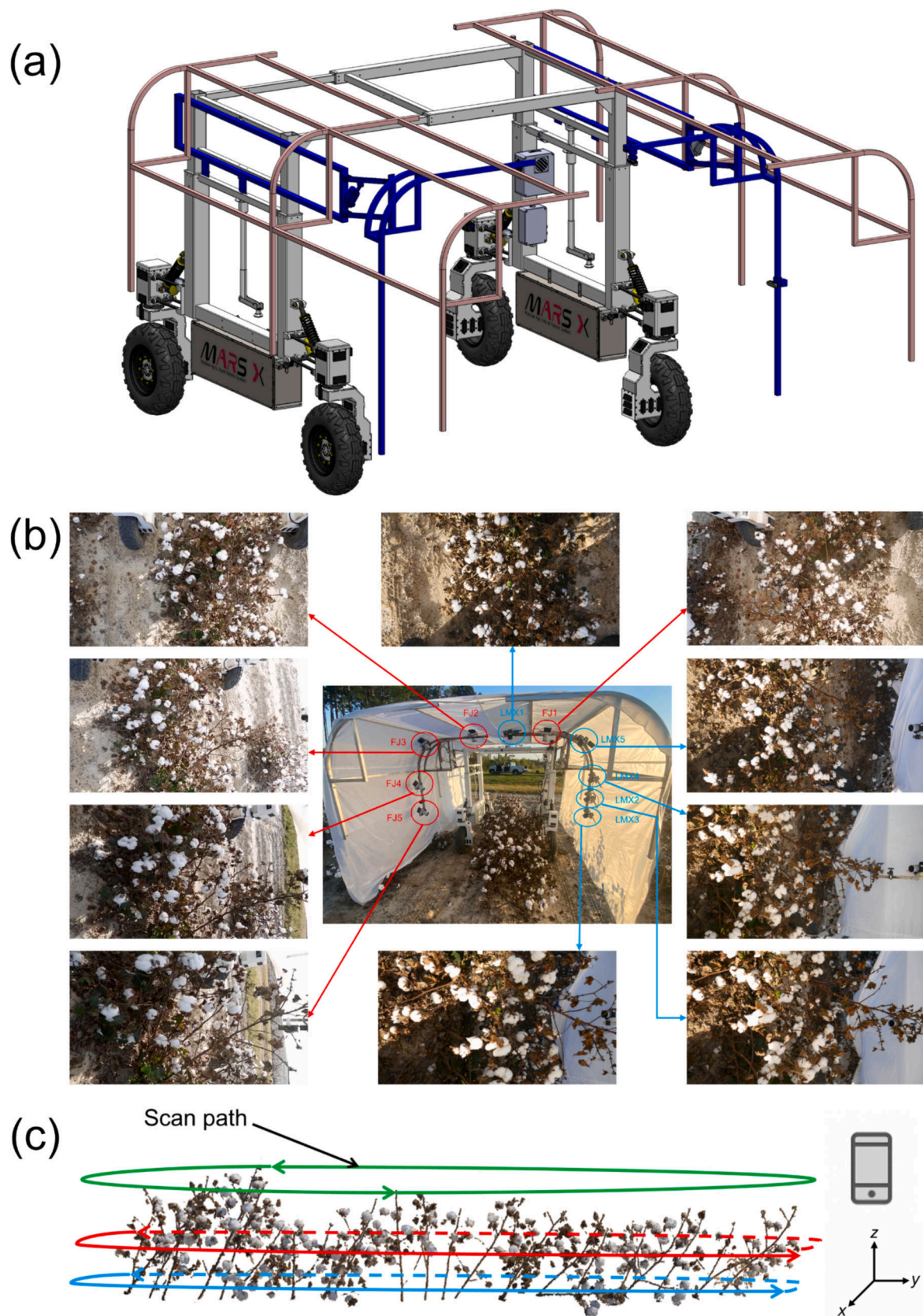


Fig. 2. Multi-view image acquisition system. (a) Simulation model of the ground robot MARS. (b) The MARS robot is equipped with 10 cameras, with five Lumix cameras (LMX) marked by blue circles and five Fuji cameras (FJ) marked by red circles. (c) Data collection path using a smartphone. The green line represents the top view, while the red and blue lines indicate side views at different heights. Data were collected at three height levels: upper, middle, and lower.

Table 1

Summary of data acquisition devices and parameters. Both the Lumix and Fuji cameras operated in automatic mode during image acquisition.

Camera model	Lumix DMC-G7	Fuji X-A10	iPhone 11
Lens model	14 mm f/3.5–5.6	16 mm f/3.5–5.6	13 mm f/2.4
Resolution (pixels)	3840 × 2160	1920 × 1080	3840 × 2160
Frame rate (FPS)	30	30	30
Video duration (s)	40	40	130

sampling strategy provided sufficient view redundancy for COLMAP-based camera pose estimation and NeRF reconstruction. The extracted overlapping multi-view images were pre-processed using COLMAP, which performed feature extraction, feature matching, and 3D sparse reconstruction. This process estimated the intrinsic and extrinsic camera parameters while reconstructing a sparse point cloud of the scene. Three-dimensional (3D) reconstruction and evaluation of plot-level cotton plants were conducted using Nerfstudio software (version 1.1.4; <https://docs.nerf.studio/>). Based on the RGB images and estimated camera parameters, the Nerfacto (Tancik et al., 2023) model was utilized to learn a 3D representation of the scene. Each scene was trained for 30,000 iterations to complete the reconstruction process.

2.3. Plot-level 3D cotton boll segmentation and counting

This study proposed a 2D-to-3D mapping-based workflow for plot-level cotton boll counting, effectively addressing the challenge of limited 3D annotation data. First, a SAM-based semi-automatic data labeling method was designed to improve annotation efficiency. The labeled data were used to train the YOLOv11x segmentation model, which was then used to generate 2D masks of cotton bolls from RGB images. Then, FruitNeRF was trained using the RGB images and their corresponding binary masks to incorporate cotton boll semantic information into the NeRF model, enabling the extraction of cotton boll point clouds. Finally, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) was applied to the point cloud for clustering and boll counting.

2.3.1. A cotton boll 2D mask annotation method based on a foundation model

This study developed a semi-automatic 2D mask annotation method for cotton bolls based on the Segment Anything Model (SAM; Kirillov et al., 2023). First, a YOLOv11x object detection model was trained on a cotton boll dataset collected in 2022 using a top-view acquisition method. This dataset, developed in previous research (Tan et al., 2024), had been annotated with 2D bounding boxes for each cotton boll. The

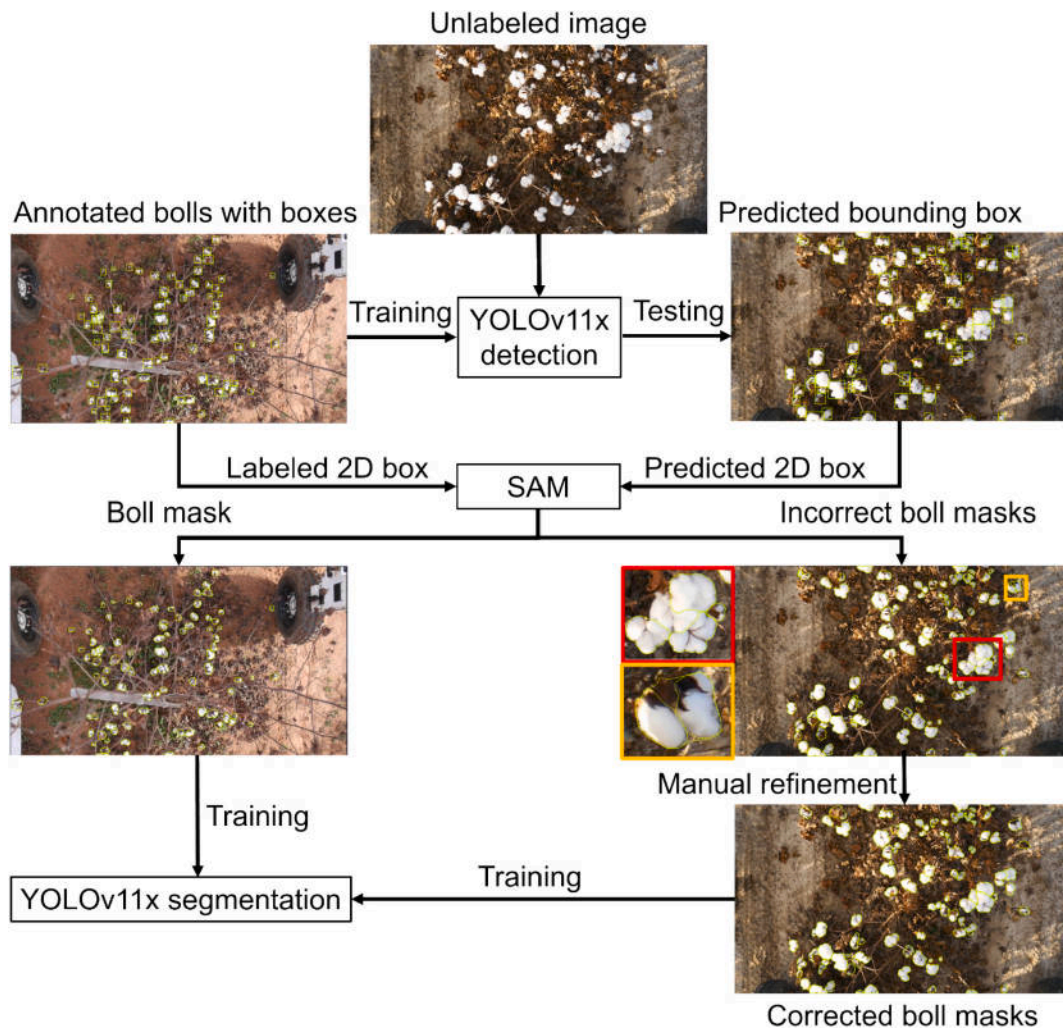


Fig. 3. Semi-automatic annotation of the cotton boll segmentation dataset and model training. The 2022 detection dataset is first used to train the YOLOv11x model, which is then applied to detect bolls in the 2024 dataset. Predicted bounding boxes from 2024 and labeled boxes from 2022 are used as prompts for SAM to generate 2D masks. Incorrect masks are manually refined to produce the final segmentation datasets for both years. Finally, a YOLOv11x segmentation model is trained on the 2022 dataset and fine-tuned with the 2024 data.

trained model was then applied to new, unlabeled data to detect cotton bolls and generate corresponding 2D bounding boxes (Fig. 3). YOLO (You Only Look Once) is an end-to-end single-stage deep learning model that has been widely applied in the agricultural domain. It features a well-developed ecosystem, a low entry barrier, simple installation and configuration, lightweight architecture, and ease of deployment. Moreover, YOLO has strong extensibility—it can not only perform object detection to generate 2D bounding boxes, but also support instance segmentation to produce 2D masks. These advantages make it particularly well suited for our task. YOLOv11x, an advanced version of the YOLO model, was employed in this section to generate 2D bounding boxes of cotton bolls for data annotation.

Next, the predicted 2D bounding boxes of cotton bolls were provided as prompts to the SAM model (sam_vit_h_4b8939.pth) to generate instance-level 2D masks of the cotton bolls. As a foundation model with zero-shot segmentation ability, SAM diverges from conventional supervised methods by using prompts—such as bounding boxes, points, or text—to initiate segmentation, eliminating the need for dense manual labels. Trained on a large-scale dataset containing over 11 million images and more than 1 billion masks, SAM demonstrates strong generalization to unseen tasks and objects. This extensive training enables SAM to perform accurate segmentation without requiring task-specific fine-tuning. For the new data collected in 2024, the 2D bounding boxes predicted by YOLOv11x were used as prompts, while for the 2022 data, manually annotated 2D bounding boxes were provided as prompts to SAM to generate 2D masks of cotton bolls.

The 2D masks generated by SAM required further manual refinement. When cotton bolls were occluded by the main stem or branches, the generated masks were often incomplete and needed manual correction. YOLOv11x occasionally produced incorrect detections or missed cotton bolls entirely, leading to inaccurate or missing masks. To address these issues, the SAM-generated masks and corresponding RGB images were imported into Roboflow (<https://roboflow.com/>) for manual refinement by trained personnel. Finally, a 2D segmentation dataset of cotton bolls was obtained.

The 2D segmentation dataset of cotton bolls was used to train the YOLOv11x segmentation model, which was subsequently applied to generate cotton boll masks from the images. Table 2 presents a summary of the cotton boll segmentation dataset. In this study, the YOLOv11x model was first trained on the 2022 dataset and then fine-tuned on the 2024 dataset to obtain the final cotton boll segmentation model. Data augmentation was applied to the training dataset, primarily including rotation, brightness adjustment, exposure, and blur augmentation. For validation set construction, images from all viewpoints were included.

2.3.2. Point cloud semantic segmentation based on FruitNeRF

FruitNeRF (Meyer et al., 2024) is a unified fruit segmentation framework that leverages NeRF to construct a 3D representation of the scene and incorporates semantic information for exporting the fruit point cloud. In this study, the original manual prompt-based mask generation process was replaced with an automated workflow using YOLOv11x to generate 2D cotton boll masks for semantic supervision. The model first takes an unordered set of RGB images and generates semantic segmentation masks for the fruits in each image. In the workflow of this study, YOLOv11x was used to obtain 2D masks of cotton bolls. The RGB images and the corresponding binary masks of the cotton bolls were combined to train FruitNeRF, enabling the entire scene to be enriched with semantic information of the fruits (Fig. 4). The trained FruitNeRF was then uniformly sampled to generate the cotton

boll point cloud. FruitNeRF also includes a denoising operation for the segmented cotton boll point cloud, which filters out points that lack the minimum number of neighboring points within a specified radius.

FruitNeRF extends the original NeRF framework by introducing a semantic field, in addition to the existing density and appearance fields. Inspired by Semantic-NeRF (Zhi et al., 2021), this semantic field encodes fruit-related information within the 3D scene, allowing each spatial location to be identified and labeled as either a cotton boll or background. It is implemented using a multi-layer perceptron (MLP) that approximates the semantic logits $S(r)$ for each pixel based on its 3D position. To estimate the semantic logits $S(r)$ for a pixel, volume density and semantic data are integrated over K sample points along the ray r using numerical methods.

$$S(r) = \sum_{k=1}^K T(t_k) \alpha(\sigma(t_k) \delta_k) s(t_k) \quad (1)$$

$T(t_k)$ represents the accumulated transmittance from t_n to t_k . $\delta_k = t_{k+1} - t_k$ represents the distance between adjacent sampled points. $\sigma(t_k)$ represents the volume density at the point t_k . $\alpha(\sigma(t_k) \delta_k)$ represents the light absorption value at the sampled point, calculated as: $\alpha(x) = 1 - \exp(-x)$. $s(t_k)$ represents the semantic value at the sampled point t_k , which is obtained through the MLP network and represents the semantic information of that location.

The cotton boll semantic field consists of 3 layers, with a hidden layer size of 128 neurons. The input layer has a dimension of 30, derived from a 30-dimensional latent vector from the density field, and the output dimension of the neural network is 64. This output serves as input for the neural segmentation head, which further reduces the dimension to a single class. When exporting the cotton boll point cloud, the number of rays to be sampled per batch during the point cloud generation process is set to 3000, and the number of points to be sampled along each edge of the 3D scene (bounding box) is set to 6000. Afterward, the cotton boll point cloud for a single plot is manually cropped to include only the cotton boll points of the plot of interest.

2.3.3. Clustering cotton boll point clouds for counting

The cotton boll points for each plot were obtained in the previous step. To achieve boll counting, instance segmentation was performed using a clustering method. Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) is a density-based clustering algorithm that effectively identifies clusters of arbitrary shapes and varying densities, while also handling noise points (McInnes et al., 2017). The core idea of HDBSCAN is to construct a density-based hierarchical clustering tree and extract the optimal clustering result based on criteria such as cluster stability. The algorithm relies on two key concepts: utilizing mutual reachability distance to build a dense connectivity graph, and constructing a Minimum Spanning Tree (MST) to capture the connectivity and evolution of clusters, which is then pruned to obtain the optimal clustering outcome.

The HDBSCAN algorithm consists of five key steps.

Step 1: Core distance calculation, which measures the local density around each data point and serves as the basis for constructing mutual reachability distances. Given a parameter min_samples , the core distance of a point p is defined as the distance to its min_samples -th nearest neighbor.

$$\text{core}_k(p) = \text{distance to the } k^{\text{th}} \text{ nearest neighbor of } p \quad (2)$$

Step 2: Mutual Reachability Distance Calculation. Mutual reachability distance makes points within dense regions appear closer to each other, while increasing the apparent distance between points in sparse regions. This helps distinguish clusters with varying densities.

$$d_{\text{mreach-}k}(p, q) = \max\{\text{core}_k(p), \text{core}_k(q), d(p, q)\} \quad (3)$$

Here, $d_{\text{mreach-}k}(p, q)$ denotes the mutual reachability distance, and $d(p, q)$

Table 2
Summary of cotton boll segmentation dataset.

	Training images	Validation images	Testing images
2022 data	606	260	–
2024 data	85	25	15

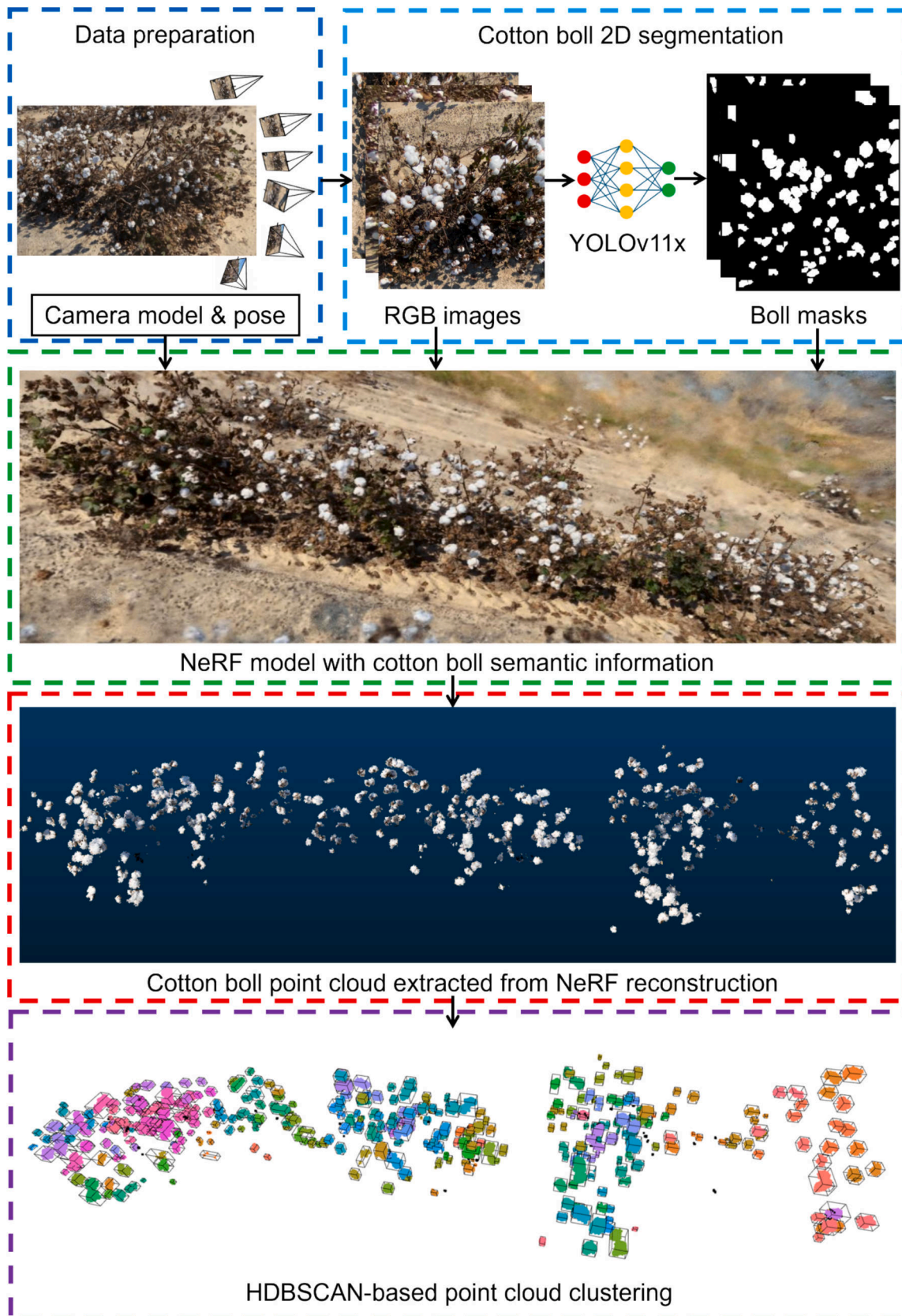


Fig. 4. Plot-level cotton boll point cloud segmentation workflow. First, camera parameters were estimated by Structure-from-Motion (SfM) using multi-view images. Then, YOLOv11x was employed to extract 2D masks of cotton bolls. Next, the posed RGB images and corresponding semantic masks were used to train a Neural Radiance Field (NeRF). A dense cotton boll point cloud was generated by uniformly sampling the NeRF model with embedded cotton boll semantic information. Finally, cotton boll instance segmentation was achieved by clustering the point cloud using HDBSCAN.

denotes the Euclidean distance between two sample points (Fig. 5).

Step 3: Build the Minimum Spanning Tree (MST). This step reveals the connectivity between data points and the underlying cluster structure, serving as the foundation for hierarchical clustering. In the graph, each node represents a sample point, and the edge weights correspond to the mutual reachability distances between point pairs. The MST is constructed using Prim's algorithm.

Step 4: Build the cluster hierarchy. First, the edges in the MST, weighted by mutual reachability distances, are sorted in ascending order. Then, the upper limit of mutual reachability distance is set as the density threshold and is gradually increased. At each threshold level, all edges with weights exceeding the current threshold are removed, leading to the disconnection of the graph. Each disconnection may correspond to a potential cluster split. To avoid clustering small groups of points into separate clusters, HDBSCAN introduces the `min_cluster_size` parameter to filter newly formed connected components, discarding those whose number of points falls below the specified threshold.

Step 5: Extract the optimal clusters from the hierarchical clustering tree based on cluster stability. $\lambda = 1/\text{mutual reachability distance}$ represents the density. $\lambda_{\text{start}}(p)$: the density at which point p joins the cluster; $\lambda_{\text{end}}(p)$: the density at which point p leaves the cluster. The larger the total sum of these values across all points in a cluster, the more stable the cluster is, and the more likely it is to be retained. HDBSCAN requires two manually set hyperparameters, `min_cluster_size` and `min_samples`, which were optimized through repeated experimentation to obtain the best threshold values.

$$\text{Stability}(C) = \sum_{p \in C} (\lambda_{\text{end}}(p) - \lambda_{\text{start}}(p)) \quad (4)$$

2.4. Implementation details

The 3D reconstruction of plot-level cotton fields and the segmentation tasks based on the improved FruitNeRF model were both conducted on the HiPerGator high-performance computing cluster, using a computing node equipped with 8 AMD EPYC Rome CPU cores and a single NVIDIA DGX A100 GPU (80 GB), operating under the Linux system. The YOLOv11x segmentation network was also trained on the same computational environment. The software stack comprised Python 3.10, PyTorch 1.12.1, and CUDA 11.3. Training configurations included an

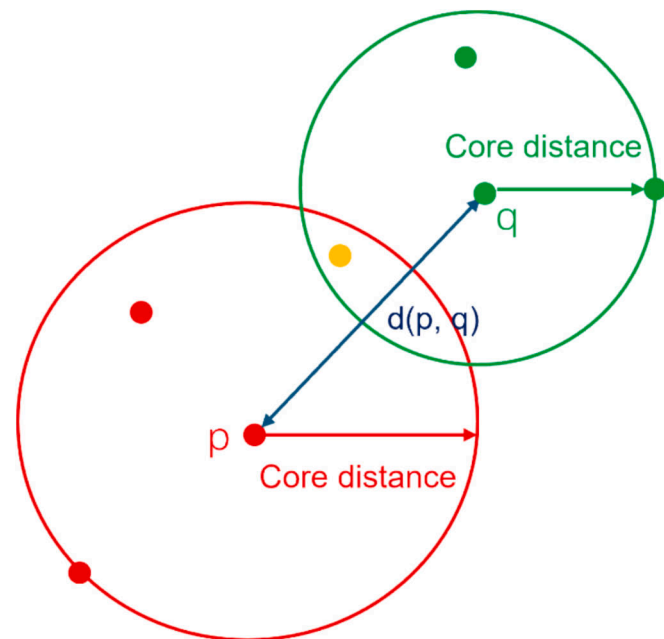


Fig. 5. Core distance and reachability distance $d(p, q)$ in HDBSCAN.

input resolution of 1280×1280 pixels, a batch size of 8, and a total of 300 training epochs. All other hyperparameters followed the default settings. (The specific dataset can be found at the link: https://figshare.com/articles/dataset/2D_labeled_data/32237358?file=64438707). For the 3D reconstruction of cotton plants, the NeRF-based experiments were conducted using the Nerfstudio-1.0.0 framework with the Nerfacto model. The model was optimized using the Rectified Adam optimizer with a learning rate of 0.0005. Training was performed for 30,000 iterations with a batch size of 1024 rays per iteration. The default loss functions provided in the Nerfacto implementation, including RGB reconstruction loss and regularization losses, were used during optimization. The FruitNeRF model was evaluated with its publicly released implementation.

2.5. Evaluation metrics

Given that the MARS robot is equipped with two different types of cameras, the image quality acquired by each was systematically evaluated. The image sharpness was quantitatively assessed using the variance of the Laplacian, which reflects the amount of high-frequency information (such as edges and textures) present in the grayscale image. The sharpness score S is computed as shown in Eq. (5). A higher value of S indicates a sharper image with more pronounced edges. During evaluation, the final score was obtained by averaging the scores of 10 images.

$$S = \text{Var}(\nabla^2 G) = \frac{1}{N} \sum_{i=1}^N (\nabla^2 G(i, j) - \mu)^2 \quad (5)$$

where $\nabla^2 G$ denotes the Laplacian response at pixel (i, j) of the grayscale image G , μ is the mean of all Laplacian responses, and N is the total number of pixels in the image.

The Mean Squared Error (MSE) was adopted to quantify the difference between two images. A lower MSE value indicates a higher level of similarity. In this context, m and n denote the height and width of the image, respectively. $I(i, j)$ represents the pixel intensity at position (i, j) in the original image, while $K(i, j)$ corresponds to the pixel intensity at the same position in the processed image. The Peak Signal-to-Noise Ratio (PSNR), a widely used metric for assessing image quality, reflects how closely the processed image matches the reference image. A higher PSNR value signifies better image quality (Eq. (7)). PSNR is computed based on the MSE, where MAX stands for the maximum possible pixel value in image I .

$$\text{MSE} = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (6)$$

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) \quad (7)$$

The Structural Similarity Index (SSIM) was employed to evaluate image similarity in terms of brightness, contrast, and structural information. In this metric, μ_x and μ_y represent the mean pixel intensities of images x and y , respectively, while σ_x and σ_y denote their variances. σ_{xy} refers to the covariance between the two images. The constants c_1 , c_2 , and c_3 are used to stabilize the calculations of brightness, contrast, and structure components. In addition, the Learned Perceptual Image Patch Similarity (LPIPS), also known as perceptual loss (Zhang et al., 2018), compares the deep features of image patches, aiming to reflect perceptual similarity as captured by the human visual system.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)(\sigma_x + c_3)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)(\sigma_x\sigma_y + c_3)} \quad (8)$$

The accuracy of cotton boll number estimation was evaluated using root mean square error (RMSE), mean absolute error (MAE), and mean ab-

solute percentage error (MAPE). In these formulas, N_i denotes the predicted count for the i -th sample, m_i denotes the corresponding ground truth, and n represents the total number of evaluated samples.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (N_i - m_i)^2} \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |N_i - m_i| \quad (10)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{N_i - m_i}{m_i} \right| \quad (11)$$

3. Results

3.1. 3D reconstruction result evaluation

Three-dimensional reconstruction can be successfully performed using 2D images captured by various camera types and configurations. In this study, data collection was conducted using two different types of cameras on the ground robot MARS. The performance of 3D reconstruction under different camera configurations was then compared. Given the inherently complex and dynamic nature of field environments, the reconstruction quality exhibited variability among different plots. To address this, we selected ten representative plots and calculated the average performance metrics. The evaluation results indicate that while NeRF-based 3D reconstruction can be performed using data from all camera configurations, the reconstruction quality varied substantially depending on the camera setup (Table 4). Among the tested configurations, the five Lumix cameras produced the best 3D reconstruction metrics. The smartphone also delivered promising results, particularly in terms of perceptual similarity (LPIPS). In contrast, the use of five Fuji cameras or a mixed setup combining two different types of cameras (10 cameras: five Lumix and five Fuji), resulted in lower reconstruction performance across all evaluation metrics.

It should be noted that PSNR, SSIM, and LPIPS were used in this study as relative image-level metrics to compare different image acquisition configurations under the same reconstruction pipeline. In outdoor plot-level cotton scenes, these metrics are strongly influenced by complex canopy structures, repetitive textures, partial occlusions, shadows, illumination variation, and slight plant movement during image acquisition. Therefore, the relatively modest PSNR and SSIM values observed in this study should be interpreted in the context of field conditions. The practical usefulness of the reconstructed models was further validated through downstream cotton boll segmentation and counting performance, rather than by the rendering metrics alone.

Image quality affects the performance of NeRF-based 3D reconstruction. The images produced by the Fuji camera exhibited lower visual clarity (Fig. S1), which adversely impacted the reconstruction results. This is further supported by the image sharpness scores shown in Table 3, where the Fuji camera consistently received the lowest scores. The variations in camera sensor characteristics contributed to the differences in image quality. When both types of cameras were used simultaneously, the resulting point cloud exhibits inconsistent and visually chaotic color patterns (Fig. 6b). Notably, the color style of the point cloud is consistent with that of the corresponding 2D images. Quantitative evaluation of the 3D reconstruction using the ten-camera

setup shows that its PSNR is 2.691 dB lower, SSIM is 0.1117 lower, and LPIPS is 0.2006 higher than the best-performing configuration in the experiment. These results highlight the importance of maintaining sensor consistency in multi-camera data acquisition setups to ensure high-quality NeRF-based reconstruction. Therefore, the use of different types of cameras is not recommended.

3.2. 2D instance segmentation evaluation

The 2D data annotation approach based on SAM and YOLOv11x developed in this study substantially accelerated the labeling process and reduced the manual burden for new datasets. Compared with conventional manual labeling methods, this approach significantly decreased the labor required. According to our rough estimates during the annotation process, the annotation time per image was reduced by at least 50%. Furthermore, the proposed workflow is highly adaptable and can be easily extended to the annotation of other crop species or datasets, demonstrating strong flexibility and scalability.

The YOLOv11x segmentation model achieved strong performance on the newly collected 2024 dataset, demonstrating its effectiveness in accurately segmenting cotton bolls (Table 5). As shown in the segmentation results of images captured by the Lumix cameras (Fig. 7), cotton bolls are correctly segmented from different viewpoints across multiple cameras. The instance segmentation results clearly demonstrate the robustness of the YOLOv11x model under field conditions. Notably, even in the presence of cluttered backgrounds and overlapping structures such as branches and leaves, the model consistently achieves high-precision detection of cotton bolls from multiple viewpoints and successfully segments their 2D masks. The results also indicate that cotton bolls from adjacent plots were occasionally segmented due to the canopy of the MARS system being lifted by wind. However, these instances can be manually filtered out during the subsequent 3D mapping process.

3.3. 3D instance segmentation and boll counting

The workflow proposed in this study provides a robust solution for plot-level 3D segmentation of cotton bolls, effectively addressing the challenges of 2D occlusion and limited 3D annotation. The method demonstrates reliable performance on both sparse and dense cotton boll distributions, and it can accurately segment cotton bolls regardless of whether the data were captured using high-resolution Lumix cameras or consumer-grade smartphones. It should be emphasized that the cotton boll counting accuracy reported in this section was evaluated independently by comparing predicted boll numbers with manually counted ground-truth values, rather than being inferred from PSNR, SSIM, or LPIPS. By employing the MARS ground robot equipped with a canopy, the system can partially mitigate lighting issues commonly encountered in field environments. The point clouds reconstructed from Lumix camera data exhibit minimal shadows, while those reconstructed from smartphone data contain more pronounced shadow artifacts (Fig. 8). Moreover, the scanning path of the MARS robot is more stable than that of a handheld smartphone.

The final instance segmentation is highly accurate, as illustrated by qualitative results from two representative plots acquired under different conditions (Fig. 9). A key difference between the two plots lies in cotton boll density—Plot1401 features sparse bolls, whereas Plot1332 contains densely clustered bolls. Despite this difference, the proposed method accurately localizes and counts individual cotton bolls in both cases, with predicted counts closely matching ground truth. These results confirm the robustness of the method across varying boll density levels.

The optimal HDBSCAN parameters were determined by comparing different combinations of `min_cluster_size` and `min_samples`. For the data captured by the smartphone, the MAPE of cotton boll counting reached its minimum when the parameters were set to a `min_cluster_size` value of 9 and a `min_samples` value of 2. For the data captured by the five Lumix

Table 3
Image quality metrics of different camera models.

Camera model	Lumix DMC-G7	Fuji X-A10	iPhone 11
Raw resolution (pixels)	3840 × 2160	1920 × 1080	3840 × 2160
Processed resolution (pixels)	1920 × 1080	1920 × 1080	1920 × 1080
Sharpness score	1151	521	1967

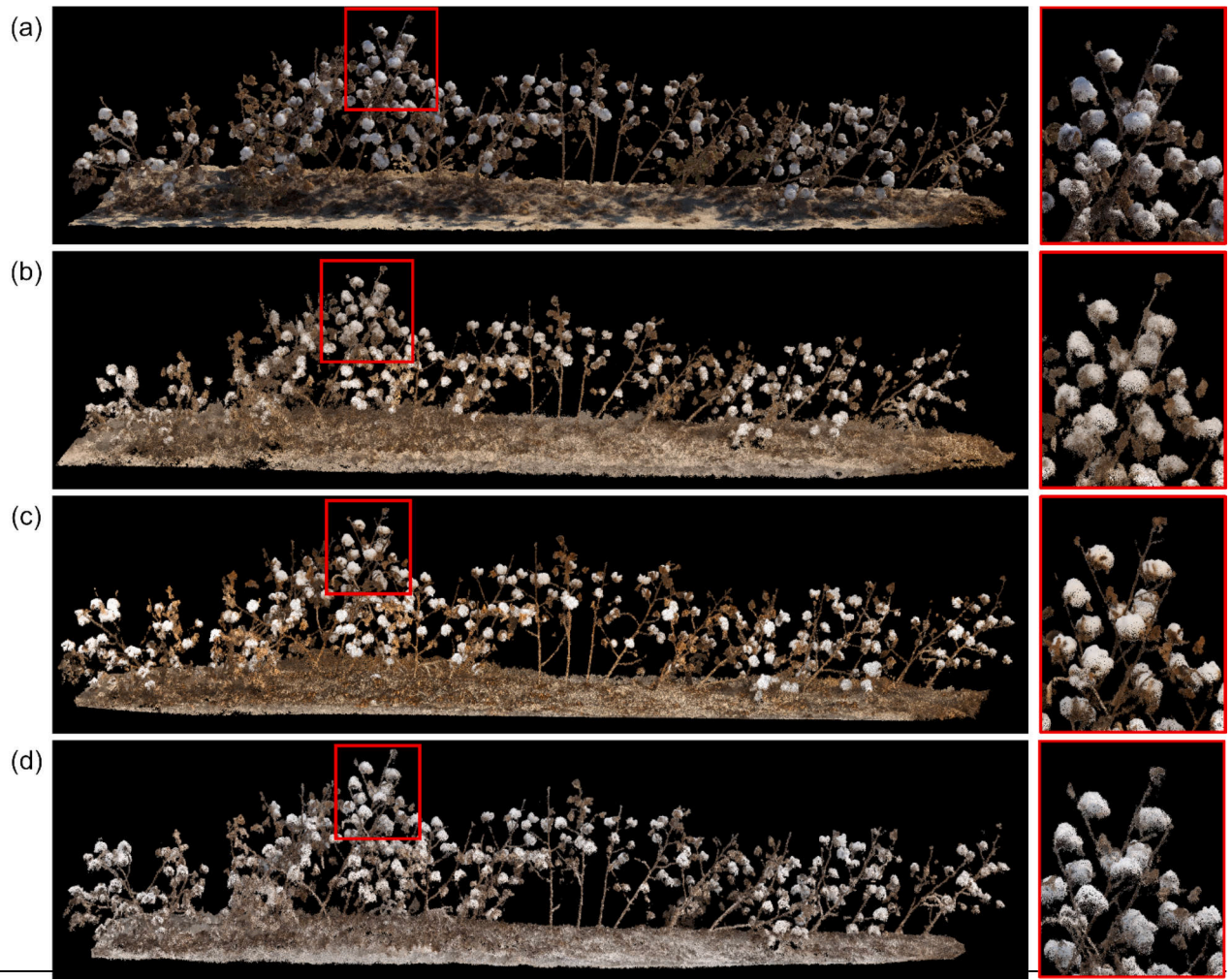


Fig. 6. Plot-level cotton plant point clouds reconstructed using different camera configurations. (a) Single smartphone. (b) Mixed setup with ten cameras of two different types (five Lumix and five Fuji). (c) Five Lumix cameras. (d) Five Fuji cameras. The regions of interest highlighted by red boxes are magnified and displayed on the right panel to show local structural details.

Table 4

Comparison of NeRF results reconstructed from images captured by different cameras. The unit of PSNR is dB.

Camera type	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
10 cameras	14.5190	0.3766	0.6915
5 Lumix cameras	17.2100	0.4883	0.5435
5 Fuji cameras	14.6100	0.3186	0.6314
Smartphone	16.2105	0.3435	0.4909

Table 5

2D instance segmentation with YOLOv11x.

2024 data	P	R	mAP50	mAP50-95
Validation	0.801	0.765	0.833	0.566
Test	0.860	0.820	0.874	0.610

cameras, the best result was achieved with a min_cluster_size value of 19 and a min_samples value of 2. These settings produced the most accurate results. However, for different breeding requirements—such as when a MAPE below 10% or 20% is acceptable—a broader range of parameter choices becomes feasible. The heatmap in Fig. S2 illustrates the results for each parameter combination, and the optimal configuration was selected in this study for yield estimation.

The proposed workflow demonstrates strong potential for accurate plot-level cotton yield estimation when integrated with both smartphone-based and Lumix-based image acquisition. The results show that boll number and yield predictions exhibited high accuracy and strong correlations with ground truth measurements, even under field environments with variable lighting and structural complexity. For the Lumix camera data, the regression between predicted and actual boll numbers (Fig. 10b) achieved a high correlation ($R^2 = 0.95$), and the corresponding yield prediction (Fig. 11b) also maintained a strong relationship with ground truth ($R^2 = 0.89$). On average, the yield estimation error per plot was approximately 148 g for both data sources. These findings highlight the practicality and adaptability of the proposed method for plot-level cotton yield estimation under real-world field conditions. Although the smartphone-based data achieved comparable boll counting and yield estimation performance, its reliance on manual data acquisition may limit its scalability in large-scale applications. By contrast, the Lumix camera integrated with the ground robotic platform provides more standardized and repeatable image acquisition, which is beneficial for scalable plot-level phenotyping. To further indicate the uncertainty of the estimation results, the key error metrics were reported with 95% confidence intervals (CIs). The CIs were estimated using bootstrap resampling with 10,000 iterations based on plot-level samples. As shown in Table 6, both smartphone-based and Lumix-based data achieved comparable estimation performance. The reported CIs provide additional information on the uncertainty of boll



Fig. 7. Visualization of cotton boll 2D instance segmentation using YOLOv11x. The left column shows the original images captured by the MARS system equipped with a Lumix camera, corresponding to Lumix 1 through Lumix 5 from top to bottom. The right column displays the segmented 2D masks of cotton bolls, with each mask assigned a randomly generated color.

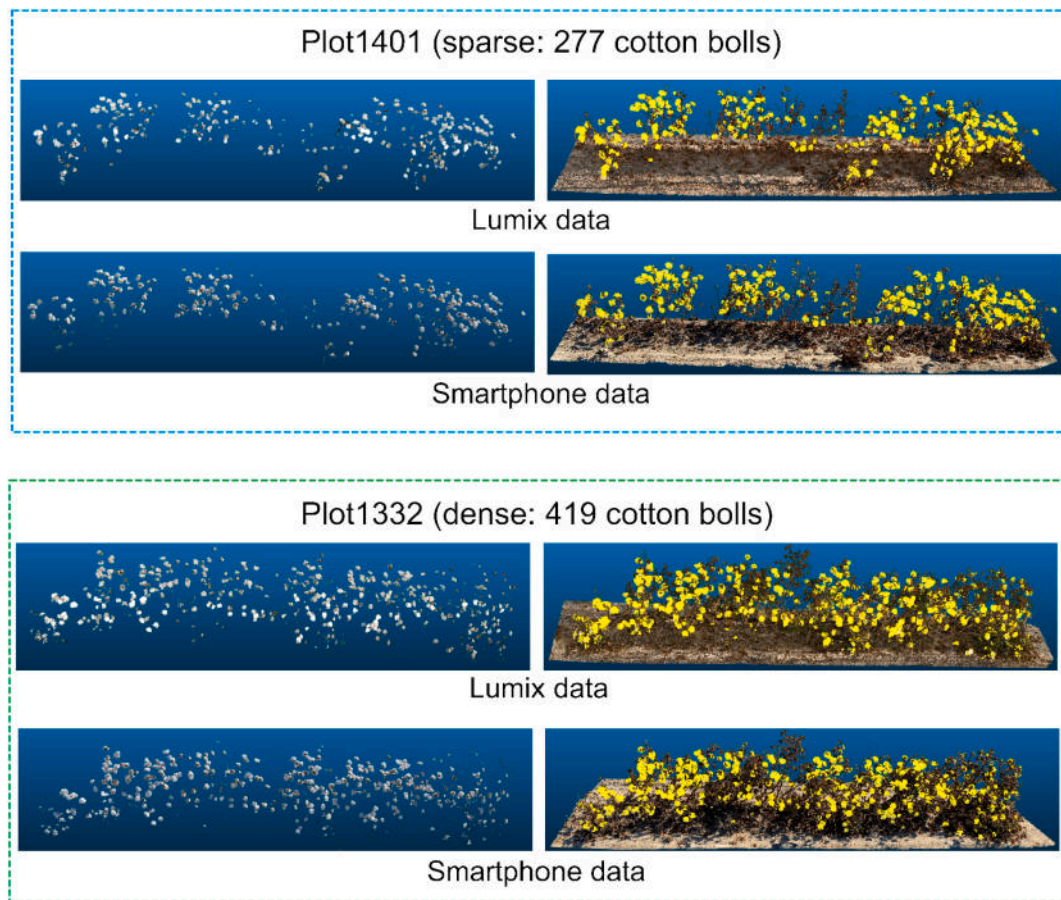


Fig. 8. Semantic segmentation results of cotton bolls using FruitNeRF. Two representative plots with different boll densities are shown, captured using different imaging devices. The left column shows the segmented bolls only, and the right column shows the bolls mapped back onto the original 3D plant models, with bolls being highlighted in yellow.

number and yield estimation across plot-level samples.

To enhance visualization, the yield information for each plot was presented in the form of a heatmap (Fig. 12). Although the predictions from the two imaging systems differ slightly, both show strong spatial agreement with the measured yield distribution. Plots with high predicted boll numbers (e.g., row 1, column 2 and row 2, column 2) are highly consistent with plots with high boll weights. Conversely, plots with low predicted numbers are consistent with low yield areas (e.g., row 2, column 7). These results demonstrate the potential of the robot-assisted Lumix-based reconstruction workflow for spatially resolved boll number estimation and yield mapping. The color gradient on the three maps further highlights the agreement between the predicted numbers and actual yields, supporting the feasibility of using 3D point cloud segmentation for field yield estimation.

4. Discussion

Two custom-designed data collection methods were employed in this study to support plot-level 3D reconstruction of cotton plants using NeRF models and to export colored point clouds. The primary method utilizes a custom-developed ground robotic platform, MARS, which is equipped with multiple cameras to enable large-scale, high-throughput data acquisition in the field. This automated system significantly enhances operational efficiency and data consistency, making it particularly suitable for scalable phenotyping tasks (Table 7). In contrast, the smartphone-based data collection method offers a low-cost and user-friendly alternative, ideal for flexible field sampling scenarios. However, it also presents several practical limitations. Data collection using smartphones relies on manual operation, which demands substantial

labor and can introduce inconsistencies—for example, variations in camera angles and movement paths during the acquisition process. Moreover, smartphones are susceptible to overheating and shutdown in high-temperature environments, and their limited battery capacity and storage capacity pose further constraints for large-scale or long-duration data acquisition. Compared to manual cotton boll counting under high-temperature outdoor conditions, the robot-assisted high-throughput plot-level 3D reconstruction workflow offers a more efficient, less labor-intensive, and more comfortable alternative. The ability to generate usable plot-level 3D architectural representations and cotton boll counts directly from RGB image inputs further demonstrates the practical feasibility of the proposed workflow for field phenotyping applications. The outdoor, plot-level data acquisition method adopted in this study not only offers broader applicability than indoor or greenhouse methods (Liu et al., 2025; Xie et al., 2025), but also better reflects plant growth under natural conditions, improving the ecological relevance of the collected phenotypic traits (Song et al., 2025).

Although smartphone-based acquisition achieved competitive reconstruction and counting performance in this study, the primary advantage of the robotic platform lies in its ability to provide stable, repeatable, and standardized image acquisition for high-throughput plot-level field phenotyping. Compared with handheld acquisition, the robotic platform can follow consistent scanning trajectories, maintain more stable viewpoints, and reduce operator-dependent variability across plots and experiments. In addition, the robotic platform enables more consistent image acquisition beneath and inside dense cotton canopies, where handheld smartphone acquisition may be more limited. Therefore, the value of the robotic platform is not solely reflected by reconstruction accuracy, but also by its scalability and acquisition

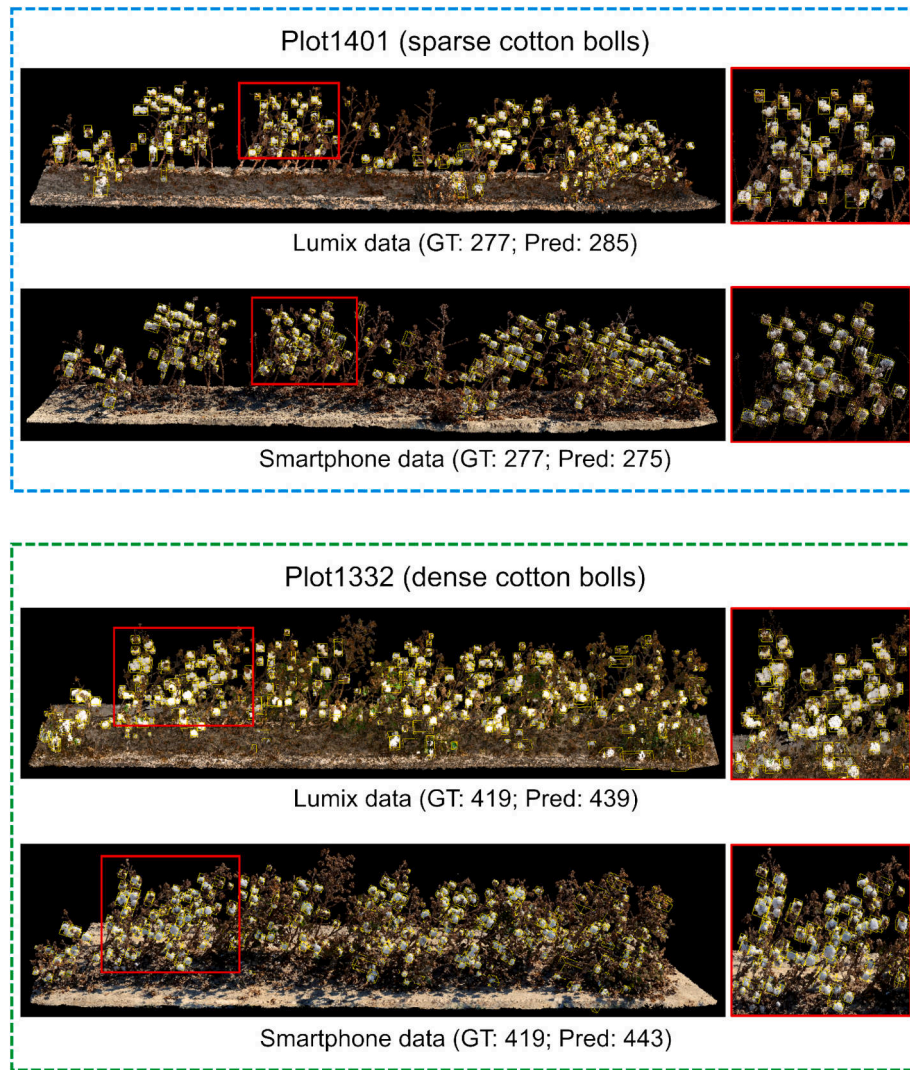


Fig. 9. Cotton boll instance segmentation results. Two representative plots with different cotton boll densities, captured using different imaging devices. Each cluster represents an individual cotton boll and is enclosed by a yellow bounding box. “GT” denotes the ground truth number of bolls, while “Pred” indicates the predicted count.

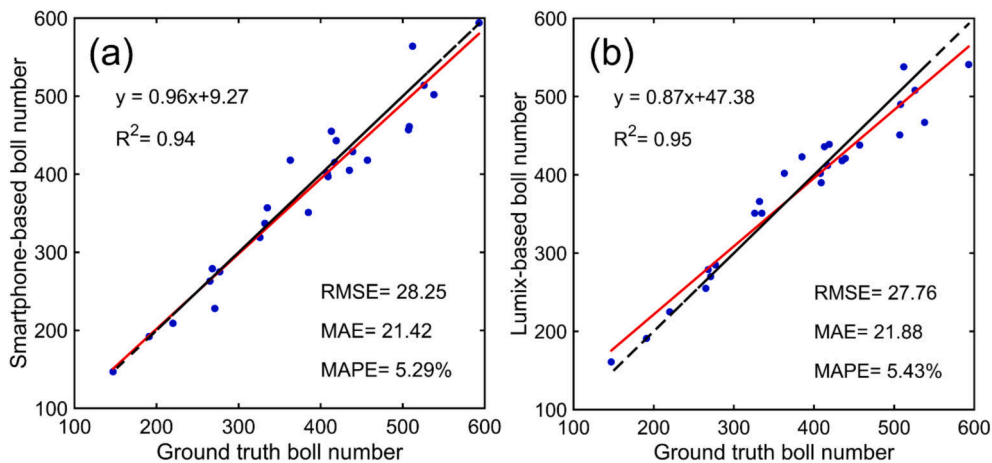


Fig. 10. Plot-level cotton boll number evaluation. Scatter plots show the relationship between predicted boll numbers and ground truth using (a) Smartphone-based and (b) Lumix-based data. Red solid lines represent regression fits, black dashed lines indicate the 1:1 reference, and blue dots correspond to individual plots.

consistency for large-scale field phenotyping applications. A quantitative comparison of viewpoint distribution and canopy spatial coverage

between robotic and handheld acquisition was not included in the current study and will be investigated in future work. In addition, the scope

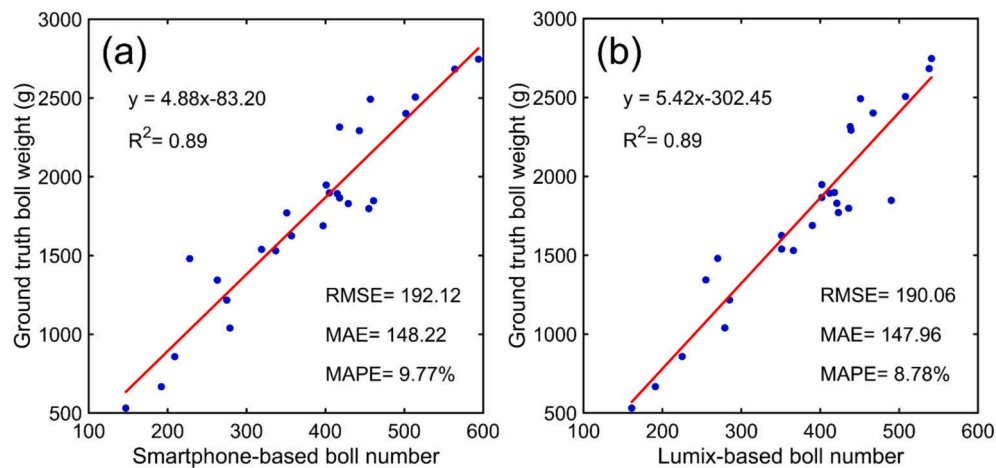


Fig. 11. Plot-level yield estimation based on predicted cotton boll number. Scatter plots show the relationship between predicted boll numbers and ground truth boll weights using smartphone (a) and Lumix (b) data. Red solid lines represent regression fits, and blue dots correspond to individual plots.

Table 6
Performance of boll number and yield estimation with 95% confidence intervals (CI). Values in parentheses represent 95% confidence intervals estimated using bootstrap resampling with 10,000 iterations. For boll number estimation, MAE and RMSE are reported in bolls. For yield estimation, MAE and RMSE are reported in grams.

		MAE (95% CI)	RMSE (95% CI)	MAPE (95% CI)
Boll number	Smartphone	21.42 bolls (14.54–28.65 bolls)	28.25 bolls (21.19–34.44 bolls)	5.29% (3.62–7.08%)
Boll number	Lumix	21.88 bolls (15.69–28.81 bolls)	27.76 bolls (19.51–35.58 bolls)	5.43% (4.08–6.80%)
Yield	Smartphone	148.22 g (93.89–195.50 g)	192.12 g (128.82–231.79 g)	9.77% (5.75–13.70%)
Yield	Lumix	147.96 g (92.07–197.85 g)	190.06 g (126.62–234.89 g)	8.78% (5.43–12.31%)

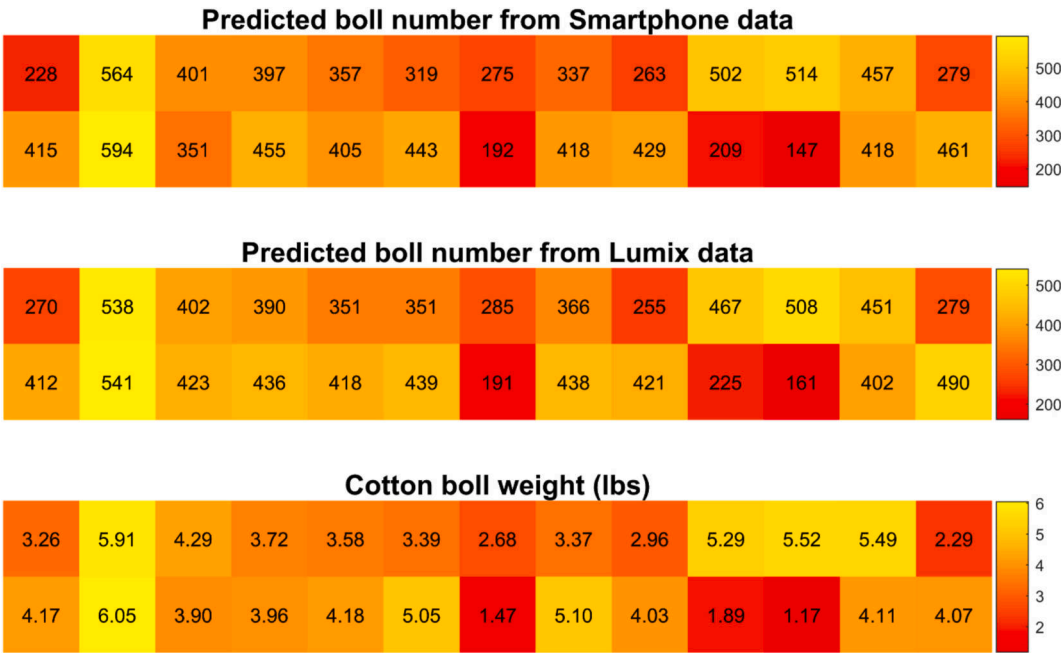


Fig. 12. Heatmaps of cotton boll yield across sampled experimental plots. From top to bottom: boll number derived from smartphone images, boll number derived from Lumix images, and boll weight per plot. Each square represents a sampled plot and the number in each square represents the boll number or weight, with spatial positions aligned one-to-one across the three heatmaps.

and objective of the present study should be further clarified. The primary objective of this study was to validate the feasibility of integrating NeRF-based reconstruction with semantic 2D-to-3D mapping for plot-level cotton phenotyping, rather than to establish a comprehensive

Table 7
Comparison of UAV-based, ground robot-based, and handheld platforms for field-based cotton phenotyping.

Platform	Cost	Resolution	Throughput
UAV	Moderate to high	Moderate; mainly captures top-view canopy information	High; suitable for rapid large-area field monitoring
Ground robot with multiple cameras	High	High; supports standardized multi-view imaging, including side-view and lower-canopy observations	Moderate to high; suitable for repeated plot-level phenotyping
Handheld smartphone	Low	High, but spatial coverage and image consistency are operator-dependent	Low; flexible but labor-intensive and less scalable

benchmark against existing 2D counting approaches. Although direct 2D counting methods generally require lower computational cost, a rigorous plot-level 2D counting baseline is not straightforward in the current field setting. A single 2D image cannot cover an entire plot, and therefore a purely 2D-based plot-level counting strategy would require either image stitching or multi-view counting with cross-view duplicate

removal (Li et al., 2025). However, image stitching in outdoor cotton field scenes is challenging because of irregular plant structures, weak and repetitive textures, occlusions, illumination variation, and possible plant movement. Multi-view 2D counting would also require reliable association of the same cotton boll across different viewpoints to avoid repeated counting. Therefore, a simple 2D baseline based on a single selected image, stitched images, or direct summation across multiple views would not provide a fair or reliable comparison with the proposed 3D workflow. In contrast, the proposed workflow enables semantic information from multiple viewpoints to be integrated into a unified 3D representation, providing semantic cotton boll point clouds suitable for downstream phenotyping analysis.

The relatively modest PSNR and SSIM values should be interpreted in the context of outdoor plot-level crop reconstruction. Unlike controlled indoor scenes or rigid objects commonly used in computer vision benchmarks, field-grown cotton plants contain highly complex, repetitive, and partially occluded structures. In addition, open cotton fibers, shadows, changing illumination, and slight canopy movement during image acquisition can further reduce view-to-view consistency. Under such conditions, image-level rendering metrics may underestimate the practical utility of the reconstructed models for downstream phenotyping tasks. In this study, PSNR, SSIM, and LPIPS were used only to

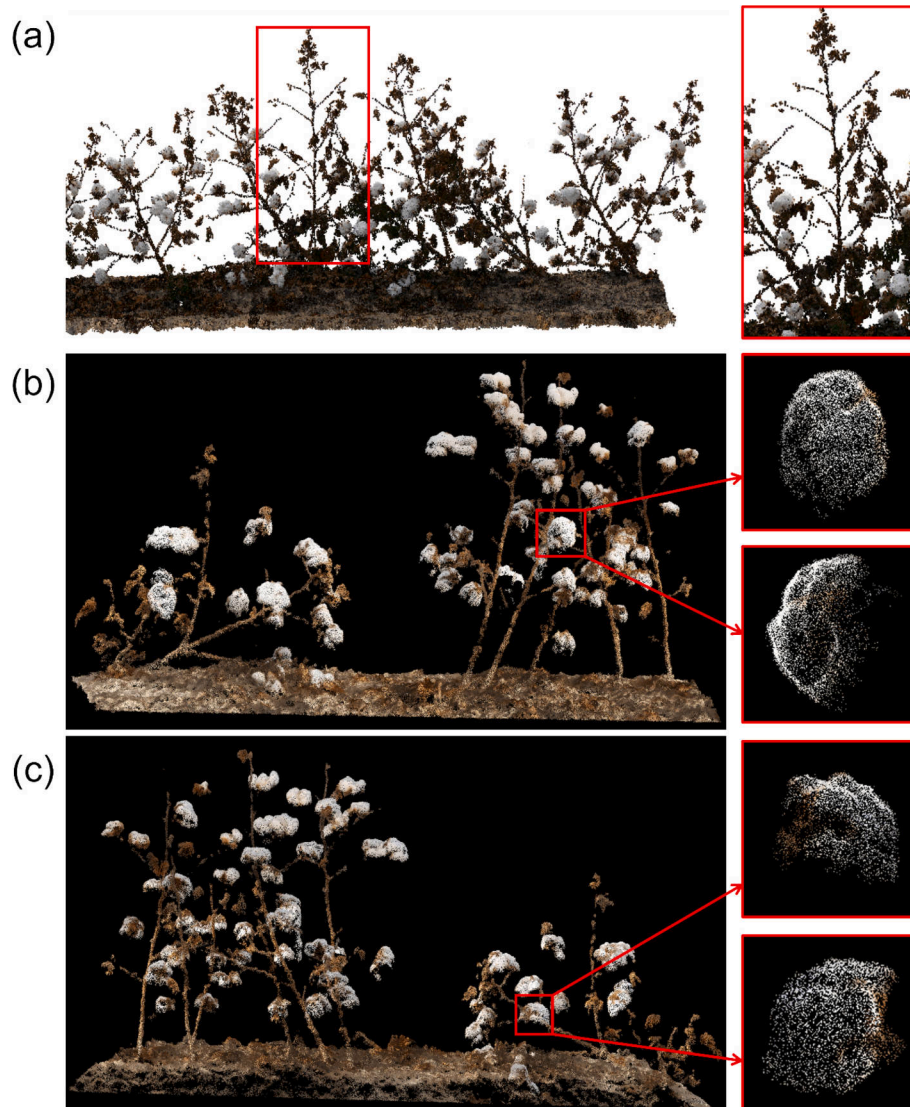


Fig. 13. More details of the point cloud obtained from data captured by the Lumix camera. (a) Visualization of branches in the cotton point cloud. (b) and (c) show the front and back views of the same point cloud, with the cotton bolls in the red boxes visualized from different angles.

compare different image acquisition configurations under the same reconstruction framework. The final cotton boll counting performance was evaluated independently using manually counted ground truth. Therefore, the 5.43% MAPE reflects the accuracy of the complete downstream workflow rather than a direct consequence of the PSNR or SSIM values.

NeRF is an open-source method and has been shown to achieve superior 3D reconstruction performance in recent plant phenotyping studies (James et al., 2025; Jiang et al., 2025c). Although UAVs have been used in some studies for rapid data collection, the resulting point clouds often lack structural completeness, particularly in capturing branches and the undersides of cotton bolls (Xiao et al., 2024), whereas our proposed method is capable of reconstructing the branches and fine-grained architectural traits (Fig. 13a). Other studies have explored the use of NeRF on outdoor crops with relatively simple structures, demonstrating its feasibility, but often stopping at 3D reconstruction without progressing to segmentation or trait extraction (Zhang et al., 2024). The entire workflow designed in this study enables accurate cotton boll counting.

The workflow proposed in this study enables accurate cotton boll counting and addresses the challenges associated with the complexity and labor-intensive nature of 3D cotton boll annotation, as well as the limited availability of plot-level cotton point cloud data. It offers clear advantages over direct 2D image-based counting methods by effectively overcoming occlusion issues. In boll counting across different plots, our approach achieved a MAPE of 5.29% using smartphone data and 5.43% using Lumix camera data, whereas a previous study relying solely on 2D images reported a MAPE of 8.92% (Sun et al., 2019). Compared with a previous study on 3D plot-level analysis, our method eliminates the need for manually designed features during training, operates on larger-scale plots, and produces more accurate counting results (Sun et al., 2020). Overall, the method developed in this study holds great promise for applications in cotton boll yield estimation.

The lower reconstruction performance observed in the mixed 10-camera setup should not be interpreted as evidence that fewer cameras are inherently better for NeRF-based reconstruction. Instead, the results suggest that image consistency across views is more important than simply increasing the number of cameras under the current experimental conditions. As shown in Table 3, the Lumix cameras produced higher-quality and more consistent images. In contrast, the mixed setup combined two different camera types (Lumix and Fuji), which introduced differences in image sharpness, color response, and imaging characteristics. Therefore, the mixed-camera configuration should be considered a limitation of the current data acquisition setup. In future experiments, all cameras will be standardized to the same camera type to improve consistency across views and further enhance reconstruction performance.

Although this study achieved plot-level 3D reconstruction of outdoor cotton plants, using cameras with different models and resolutions resulted in reduced point cloud quality. Consequently, only the data captured by five Lumix cameras were used for cotton boll counting. However, these five cameras provided only a side view and a top view of the cotton bolls, lacking imagery from the opposite side. This led to partially incomplete point clouds. As shown in Fig. 13 (b, c), two side views of the same point cloud are presented, and the enlarged views of the cotton bolls from different angles illustrate the missing regions. Nevertheless, this did not affect the accuracy of cotton boll counting, as the minor incompleteness of individual bolls had negligible impact. Furthermore, the ability of the proposed method to reconstruct complete plant structures highlights its potential for future phenotypic analyses of main stems and branches.

The accuracy of cotton boll counting is not only related to the segmentation precision of FruitNeRF, but also depends on the clustering results of HDBSCAN. Fig. 14 illustrates a common issue in the clustering step of the cotton boll counting pipeline. As highlighted by the red circle, two adjacent cotton bolls are incorrectly grouped into a single cluster by



Fig. 14. Limitations of cotton boll clustering results. The two cotton bolls within the red circle were grouped into one cluster.

HDBSCAN. This occurs due to the close spatial proximity and partial occlusion between the bolls, which makes it difficult for the density-based algorithm to distinguish them as separate instances. Such cases can lead to slight underestimation in the final boll count and suggest the need for further refinement in clustering parameters or post-processing strategies.

Several limitations of the current study should also be acknowledged. First, the experiments were conducted under post-defoliation conditions with relatively low planting density, which reduced canopy closure and partially alleviated severe occlusion. Although these conditions remain representative of certain breeding and phenotyping scenarios, they are less challenging than dense commercial cotton fields with fully closed canopies. It should also be emphasized that the current study primarily focused on plot-level phenotyping scenarios in cotton breeding trials, where relatively lower planting density and plot-based evaluation are commonly adopted experimental conditions. Under high-density field conditions, severe occlusion, incomplete viewpoint coverage, and reconstruction instability may substantially increase the difficulty of semantic 3D reconstruction and cotton boll separation. Therefore, the conclusions of the present study should be interpreted within the boundary conditions of the current experimental setup. Future work will investigate the robustness of the proposed workflow under denser and more complex canopy conditions.

In addition, several limitations related to the camera comparison experiments should also be noted. The datasets were collected under different field conditions and acquisition trajectories. Variations in illumination, collection timing, and scanning paths may have influenced the reconstruction and counting performance observed among different camera configurations. Therefore, the current comparisons should not be interpreted as strictly controlled evaluations of camera hardware performance alone. Future studies should perform fully controlled experiments under standardized illumination and acquisition conditions to enable more rigorous camera and sensor comparisons. Another limitation of the current workflow is its relatively high computational demand associated with NeRF optimization and semantic reconstruction. Although the proposed framework demonstrated promising performance for plot-level cotton phenotyping, the current implementation is not yet suitable for real-time or large-scale high-throughput field scouting applications.

Future work will mainly focus on two aspects. First, the data acquisition process will be further optimized by using a uniform camera

system to eliminate inconsistencies in image quality and color, thereby improving the completeness and accuracy of 3D reconstruction. A more comprehensive multi-view acquisition setup will also help reduce missing regions in the point cloud caused by limited viewpoints. Second, most existing studies on cotton plants have focused on indoor data and cotton boll counting, with limited attention given to the measurement of architectural traits (Jiang et al., 2025b). In dense commercial cotton fields with fully closed canopies, severe self-occlusion and limited visibility of interior cotton bolls may further increase the risk of incomplete reconstruction and semantic separation errors. In the future, we plan to extend our approach to quantify architectural traits of outdoor, plot-level cotton plants—such as plant height and internode length—based on the reconstructed 3D models. This will provide valuable insights for phenotypic studies beyond yield estimation.

5. Conclusions

This study proposes a ground robot-assisted high-throughput data acquisition system and establishes a novel workflow for NeRF-based plot-level 3D reconstruction of cotton plants, boll counting, and yield estimation in the field. The proposed workflow demonstrates the feasibility of generating semantic plot-level 3D cotton models and performing cotton boll counting directly from RGB image inputs collected under real field conditions. By obtaining high-quality 2D masks of cotton bolls from RGB images and projecting them onto the 3D point cloud using FruitNeRF, the method effectively bridges 2D segmentation with 3D analysis to achieve plot-level boll counting and yield estimation. A comparative analysis showed that the data collected by the Lumix camera and the smartphone resulted in comparable cotton boll counting accuracy; however, the Lumix-camera configuration integrated with the MARS robotic platform was better suited for high-throughput field deployment because it provided more stable, repeatable, and standardized image acquisition. The proposed approach offers a valuable solution for high-resolution plant mapping and 3D phenotyping, and it is adaptable to a wide range of crops. In future work, we will adopt a standardized setup using ten identical cameras for data acquisition and focus on extracting plant architectural traits from field data, such as branch angles and main stem height at the plot level.

Author contributions

L.J. and C.L. conceived the idea and designed the experiments. P.W. C. contributed to the preparation of materials. L.J. conducted the experiments and analyzed the data. L.J., C.L., L.F., and P.W.C. interpreted the results and wrote the manuscript. All authors read and approved the final version of the manuscript.

CRedit authorship contribution statement

Lizhi Jiang: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Peng W. Chee:** Writing – review & editing, Supervision, Resources. **Changying Li:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition. **Longsheng Fu:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2026.112063>.

Data availability

Data will be made available on request.

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